

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Igneous Petrology L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus: Igneous rocks-their classification, field occurrence, textures; Internal structure of the Earth, geothermal gradient, mantle melting mechanisms, generation of basaltic magmas; Magma generation, differentiation, melt properties, use of binary and ternary systems to understand melting and crystallization behavior of rocks and magmas, application of major, trace elements, and radiogenic/stable isotopes in igneous petrogenesis; Common igneous rocks and their plate tectonic associations/petrogenesis

Reference Books:

- 1) Principles of Igneous and Metamorphic Petrology by John D. Winter
- 2) Principles of Igneous and Metamorphic Petrology by Anthony R. Philpotts and Jay J. Ague
- 3) Igneous Petrogenesis-A Global Tectonic Approach by M. Wilson

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Igneous rocks-their classification, field occurrence, textures; Internal structure of the Earth, geothermal gradient	4
2.	Mantle melting mechanisms, generation of basaltic magmas	3
3.	Magma generation, differentiation, melt properties	5
4.	Use of binary and ternary systems to understand melting and crystallization behavior of rocks and magmas	6
5.	Application of major, trace elements, and radiogenic/stable isotopes in igneous petrogenesis	8
6.	Common igneous rocks and their plate tectonic associations/petrogenesis	8
Total number of hours		34